Organochlorine pesticide residues in bamboo shoot.

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BACKGROUND: Xenobiotic organochlorine pesticides (OCPs) are a major environmental problem because of their historic widespread use, pronounced persistence against chemical and biological degradation, and bioaccumulation in the food chain. Pesticide use is prevalent in the production of edible bamboo shoots, which are exported widely from China. To evaluate the quality of Chinese bamboo shoots we determined the residual content of some OCPs in shoot samples. RESULTS: Three types of OCPs-hexachlorocyclohexane (HCH), 1,1,1-trichlor-2,2-bis(p-chlorophenyl)ethane (DDT) and pentachloronitrobenzene (PCNB)-were detected in bamboo shoots from Zhejiang province, China. Detection rates were 100%, 100% and 75% for HCH, DDT and PCNB, respectively. However, the average residue concentration did not exceed the maximum residue limit for pesticides detected in food in China (50 µg kg(-1)). In terms of residue concentrations of the pesticides, 82.14% of the bamboo shoot samples could be classified as safe. CONCLUSION: While all sampled bamboo shoots contained OCP, most (82.14%) were safe for consumption. 2010 Society of Chemical Industry.